Und2 - A Diablo 2 mod for quality of life and extending the fun duration of the game

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1 Goals

The mod is not aimed at necessarily modifying the game too far. Diablo 2 is a great game and most of its mechanics are intricate and have lots of factors that play into them and make them interesting. However, I feel there are a couple of things that, at least for me, make the game, specially the end-game, a bit too grindy. Similarly, I feel there are a couple of mechanics (and items) that are a bit under-used and become too useless in the end-game. The goals of this mod is to deal with both of these (related) issues at once. In particular:

- *High runes are too rare.* This is just a reality, in my opinion. Now, other mods deal with this by increasing their drop rates, but this is a lot more complicated to do and does not interact with the other goals. By using cube recipes to produce runes I can tackle both issues at once, and also modify less files.

- *Only a subset of unique/set items are useful in the endgame* And most are not useful at all. On one hand, this has the (positive) effect of dropping a unique not being automatically a nice thing, and adds another layer of mechanics to the game (finding the right unique). However, I feel this goes a bit too far. Making every unique useful in a direct way would be both extremely complicated, too much work and remove this aspect. Instead, adding **some** value to every unique incentivates MF and makes the game less about getting *that one drop*, which is nice in moderation but becomes too prevalent in endgame Diablo 2. We do this by producing another base material used in other aspects of this mod from uniques (more later).

- *The game has conceptual possibilities that cannot be exploited due to limits in itemization.* This, I feel, particularly limits the set of metagame builds in the game quite a bit. Runewords, to a degree, add dimensionality to the game with unique effects that cannot be obtained otherwise and which change this metagame. But these are still limited and extremely rare (see high runes are too rare). By adding possibilities to obtain these unique effects (aura when equipped, add skills of other classes, casts on attack, ignore defense, piercing attack, enemies flee, etc.) in more targetted and flexible ways, we open up the possibilities for builds quite a lot. We call these Augments. These Augments are one-of-a-kind effects, in the sense that stacking multiple of them is not really benefitial (on top of being limited).

- *Magic and rare items are not good enough.* Sometimes they are, but only in very specific cases. Now, it is dangerous to simply make them stronger because then they would overshadow uniques and sets. We combine this with the previous point to allow the adding of these unique effects to rares and magic items, but only in limited amounts / ways, to make magic and rare items more viable for endgame, while hopefully not making uniques completely irrelevant.

- *Good magic and rare items are (a bit) too hard to find.* The combination of affixes needed to produce a good magic or rare item is often too specific and takes large amounts of drops to obtain. The ability to reroll

magic items with perfect gems helps a bit with this, but it's still very limited. We add possibilities for rerolling both magic and rare items more easily, so while there is still a random element (that distinguishes from uniques) and it's very hard to find the perfect magic/rare item, it's easier to find a pretty good one.

- *A lot of white/magical/rare items are useless while their equivalent white/magical/rare are not.* This is a minor problem, but we add recipes to transform whites, blues and yellows into whites, blues and yellows (with random affixes in each case), add sockets more reasily, transform ethereal into non-ethereal and viceversa.

- *There is a point at which you can't really get much better.* Diablo 2 has a balance in that it's quite easy to get close to optimal, but it's extremely hard to get optimal. This works fine, except for one reason: For people who want long-term play, it becomes a grind to find that Holy Grail that will give you the last extra push forward. In this mod, I add another type of Augments specifically for charms and jewels (while allowing large amounts of sockets for non-white items), which add power-level stats that are usually only found in uniques or limited in their sources, in very small amounts that are noticeable but not game changing. Examples include damage reduced, elemental damage, reduce enemy resistance, maximum life, maximum mana, faster cast rate, faster hit recovery, crushing blow, etc. We also give rare uniques another life by making them useful for this kind of additional push that is not build-defining.

We note that all of these changes are achieved almost exclusively through new Horadric Cube recipes, with a couple of exceptions. This makes the mod simple and its effects limited.

2 Runes

First, we slightly tweak the runes treasure classes to make high runes slightly more common in drops. This should also make farming for runes slightly more worth it.

However, in order to make high runes more accessible, the main thing we do is that we add cube recipes that allow us to obtain them more easily. The basic cube recipes to obtain higher runes have the problem of being exponential (requiring multiple of each previous layer to produce). Even with small

numbers, this becomes very quickly insufficient (most people never cube runes above Ort). The new recipes instead upgrade a rune to the next one using additional materials that are generally obtainable through similar means to runes themselves. In particular, Perfect Gems and Unique/Set Essences (see the section for Uniques/Sets). In particular, we add the following recipes:

Base rune	**Added materials**	**Result rune**
El	1 Perfect Gem	Eld
Eld	1 Perfect Gem	Tir
Tir	1 Perfect Gem	Nef
Nef	1 Perfect Gem	Eth
Eth	2 Perfect Gems	Ith
Ith	2 Perfect Gems	Tal
Tal	2 Perfect Gems	Ral
Ral	2 Perfect Gems	Ort
Ort	2 Perfect Gems	Thul
Thul	3 Perfect Gems	Amn
Amn	3 Perfect Gems	Sol
Sol	3 Perfect Gems	Shael
Shael	3 Perfect Gems	Dol
Dol	3 Perfect Gems	Hel
Hel	4 Perfect Gems	Io
Io	4 Perfect Gems	Lum
Lum	4 Perfect Gems	Ko
Ko	4 Perfect Gems	Fal
Fal	5 Perfect Gems	Lem
Lem	5 Perfect Gems	Pul
Pul	5 Perfect Gems	Um
Um	6 Perfect Gems,1 Unique Essence	Mal
Mal	6 Perfect Gems,1 Unique Essence	Ist
Ist	6 Perfect Gems,1 Unique Essence	Gul
Gul	7 Perfect Gems,2 Unique Essences	Vex
Vex	7 Perfect Gems,2 Unique Essences	Ohm
Ohm	7 Perfect Gems,2 Unique Essences	Lo
Lo	7 Perfect Gems,2 Unique Essences	Sur
Sur	8 Perfect Gems,3 Unique Essences	Ber
Ber	8 Perfect Gems,3 Unique Essences	Jah
Jah	8 Perfect Gems,3 Unique Essences	Cham
Cham	8 Perfect Gems,3 Unique Essences	Zod

To summarize, this means that one can obtain a Zod with 1 El rune, 138 Perfect Gems and 23 Unique Essences. This is still quite expensive, but enor-

mously less than it is in the basic game. It still gives rune drops (specially intermediate ones) a lot of value.

On the other hand, there is no way in Diablo 2 to *downgrade* runes, and sometimes you have runes that are higher than what you like. Similarly, we wanted a way to be able to reroll magic/rare items more easily. Runes gives us just this, using runes to reroll magics/rares while downgrading them (one step at a time, so a Zod into a Cham, all the way into an Eld into an El, and an El into nothing). This allows for quick multicasting of the cube until we get the rune/reroll that we want. It is also a good way to use extra rune drops. Now, of course higher runes take more to upgrade and have the same effect when downgrading, so it's less worthy using them for this purpose. But this is fine, this makes saving higher runes worth it to keep.

3 Uniques (and sets)

In order to make every set and unique drop valuable, we allow to distill them into essences. Set Essences for set items and Unique Essences for unique items. Set Essences can be transformed into Unique Essences on a 2:1 ratio (2 Set Essences = 1 Unique Essence). Unique Essences are also used to upgrade higher runes, and both Set and Unique Essences are used in some Augmentation recipes. Not all uniques/sets are worth the same, though, and the simplest way that we measure this is based on type and level: (this is the same for Unique and Set versions, but for Unique and Set essences respectively).

Normal Armors	1 Essence
Normal Weapons	1 Essence
Exceptional Armors	2 Essences
Exceptional Weapons	3 Essences
Elite Armors	3 Essences
Elite Weapons	5 Essences
Rings	2 Essences
Amulets	3 Essences
Charms	9 Essences
Jewels	9 Essences

However, cube recipes only allow to output 3 items. Thus, to produce more, we create Concentrated Essences and Highly Concentrated Essences, which produce 3 Essences and 3 Concentrated Essences respectively when cubed. To output 12 Essences, we output 1 Highly Concentrated Essence and 1 Concentrated Essence, for example.

Additionally, we add recipes to reroll rings and amulets, which have many unique and set versions, some of which are much more rare and interesting to the others, where a set or unique ring or amulet plus 5 Essences (of the adequate type) generates a new item of the same kind (ring/amulet and unique/set) with the same item level, but potentially a different unique or set version. This also makes unique rings, amulets, and set amulet drops a lot more valuable by themselves.

Sometimes there will be excess of set or unique essences. To this end, we add recipes to burn them inefficiently: 12 set essences is 1 random jewelry and 1 perfect gem, and 12 unique essences is 2 random jewelry and 1 perfect gem.

We also make a couple of small changes to make certain unique items or item types more interesting. First, we reduce the item level for Annihilius (unique small charm) to 94. This means that Baal, Diablo, and Nihlathak can now regularly drop it. It is still a fairly rare item, though. Similarly, we added quite a few new unique items, mostly for item types that did not have any unique before. Among these, we felt that ethereality was not entirely exploited in the game, and added powerful ethereal unique items. Some are ethereal and unsocketable, meaning that it is impossible to make them indestructible. Others are ethereal and only hold charges in powerful skills. In turn, this makes these unique items inherently temporary, while also being very rare and very powerful. These are the new uniques, all of them have some special flavour that was not really present in the game before:

Name	Base item	Attributes
Brutal Touch	Gloves	Ethereal Crushing Blow 25-50% Increased Attack Speed 50-75% 125-175% Enhanced Damage 125-175% Bonus to Attack Rating
Skillful Balance	Demonhide Sash	Ethereal +3-5 All Skills 50-100% Faster Cast Rate Increase Maximum Mana 75-150% Regenerate Mana 75-150%
Solid Step	Leather Boots	Ethereal +40-60% Faster Run/Walk +50-100 All Resistances +5-15 All Max Resistances 50-75% Faster Hit Recovery 15-30% Reduced Damage Reduce Poison Duration 50-75% +500-1000 Defense Increase Maximum Health 25-75%
Magic Fingers	Bramble Mitts	Ethereal +1 Telekinesis All Absorbs +10-50% +2 All Skills +25-50% Magic Find +20% Faster Cast Rate +20% Attack Speed +150 Mana +1-20 Lower Resist
(pending)	Crusader Gauntlets	(pending)

(pending)	Hydraskull	(pending)
(pending)	Giant Conch	(pending)
Mind Bubble	Heater	Ethereal Level 60 Energy Shield 200 Charges Level 60 Meditation Aura +300 Energy 25% chance to cast L1 Teleport when hit +75% Faster Hit Recovery
(pending)	Hyperion	(pending)
(pending)	Wyrmhide Boots	(pending)
Untouchable	Colossus Girdle	Ethereal Requirements -70-100% 100% L60 Psychic Hammer when hit L60 Summon Grizzly 200 Charges +20-30 Psychic Ward All Absorbs +10-20% +500-1000 defense +10-30 Cleansing +20-50% Faster Hit Recovery

(pending)	Wyrmhide	(pending)
(pending)	Scarab Husk	(pending)
(pending)	Diamond Mail	(pending)
(pending)	Loricated Mail	(pending)
(pending)	Great Hauberk	(pending)
The Costume	Boneweave	+11 to random barbarian skill +11 to random paladin skill +11 to random amazon skill +11 to random sorceress skill +11 to random necromancer skill +11 to random assassin skill +11 to random druid skill +11 to random warlock skill +60% faster hit recovery
(pending)	Archon Plate	(pending)
(pending)	Hellforge Plate	(pending)
(pending)	Lacquered Plate	(pending)

(pending)	Small Crescent	(pending)
The Incinerator	Feral Axe	Indestructible +1-20 Fire Mastery L10-40 Holy Fire Aura 10% L30 Enchant on hit 50% L50 Blaze on kill Cannot be Frozen Cold Absorb 50% 10-40% Increased Attack Speed Prevent Monster Heal 25% L60 Fire Golem when hit
(pending)	Spider Bow	(pending)
(pending)	Blade Bow	(pending)
(pending)	Shadow Bow	(pending)
(pending)	Great Bow	(pending)
(pending)	Diamond Bow	(pending)

(pending)	Pellet Bow	(pending)
(pending)	Gorgon Crossbow	(pending)
(pending)	Reinforced Mace	(pending)
Superluminal	Colossus Voulge	Ethereal +1-20 Lightning Mastery L60 Thunder Storm 177 Charges +1-1000 Lightning Damage 10% L25 Burst of Speed on hit +1-3 sockets +60 Fist of the Heavens Attacker Takes Lightning Damage of 150 25% L30 Lightning Sentry on kill Sunder Lightning Immunity +50 Light Radius Hit Blinds Target
(pending)	Great Poleaxe	(pending)

(pending)	Seraph Rod	(pending)
(pending)	Stygian Pike	(pending)
Unassailable Spirit	Ghost Spear	+200-400% Enhanced Damage +2%/lvl bonus to attack rating 50% L20 Heart of Wolverine when hit 20% Reanimate as: Reanimated Horde 10-20% Life Steal 50% L20 Taunt when hit L10-20 Cleansing Aura +35% Block Chance 50% Damage Taken goes to Mana
(pending)	Occult Codex	(pending)
(pending)	Forgotten Volume	(pending)
(pending)	Possessed Grimoire	(pending)
(pending)	Possessed Compendium	(pending)

(pending)	Dark Codex	(pending)
(pending)	Dark Tome	(pending)
(pending)	Grimoire	(pending)
(pending)	Compendium	(pending)
(pending)	Codex	(pending)
(pending)	Tome	(pending)
(pending)	Old Book	(pending)

(pending)	Walking Stick	(pending)
(pending)	Stalagmite	(pending)
(pending)	Shillelagh	(pending)
(pending)	Crystal Sword	(pending)
(pending)	Falcata	(pending)
(pending)	Hydra Edge	(pending)
(pending)	Conquest Sword	(pending)
(pending)	Mythical Sword	(pending)

(pending)	Highland Blade	(pending)
(pending)	Balrog Blade	(pending)
(pending)	Colossus Sword	(pending)
(pending)	Throwing Knife	(pending)
(pending)	Balanced Knife	(pending)
(pending)	Throwing Axe	(pending)
(pending)	Balanced Axe	(pending)

(pending)	War Dart	(pending)
(pending)	Hurlbat	(pending)
(pending)	Flying Knife	(pending)
(pending)	Polished Wand	(pending)
(pending)	Ghost Wand	(pending)

4 Runewords

We add a few runewords mainly aimed at incentivizing or enabling unusual playstyles and builds.

Name	Item type	Runes	Attributes
Yodel	Primal Helms	JahSurEth	+7 to Warcries +7 to War Cry 45-60% Faster Cast Rate 15-30% Faster Hit Recovery 50% Chance to cast Level 15 Taunt when Struck +80-100% Enhanced Defense +300/lvl 80 mana
Toxicity	Necromancer Heads	KoGul	+7 to Poison Dagger +5000-10000 poison bit rate over 1500 frames (approximately 60000 poison damage over 60 seconds) 100% chance to cast level 25 Venom on strike 100% chance of open wounds +250% bonus to attack rating
See Red	Druid Pelts	ChamIthZod	15% L35 Firestorm on hit 12% L35 Molten Boulder on hit 10% L35 Fissure on hit 7% L35 Volcano on hit 1% L50 Armageddon on kill L15-25 Fanatism aura +3-5 Werewolf
Dance of the Elements	Sorceress Orbs	IstShaelZod	20% L35 Frozen Orb on hit 20% L35 Nova on hit 20% L35 Hydra on hit 300% bonus to attack rate Cannot be frozen 50% L50 Energy Shield when hit +250 Energy
Betrayal	Paladin Shields	IstLemGul	10% L35 Raise Skeleton on kill 20% L35 Poison Explosion on kill 10% L35 Grim Ward on kill 20% L35 Apocalypse on kill 3% Reanimate as Duriel +100% Poison Skill Damage +255-511 Magic Damage

Molten Core	Armor	VexEthVexRal	+1 to Fire Mastery 1% L50 Armageddon on kill Level 15-30 Holy Fire Aura 5-15% Fire Absorb +200-500 Fire Damage Regenerate Mana 50% Replenish Life +50
Frozen Core	Armor	OhmElOhmThul	+1 to Cold Mastery 1% L50 Hurricane on kill Level 15-30 Holy Freeze Aura 5-15% Cold Absorb +200-500 Cold Damage +1500 Defense +25% chance to block
Galvanized Core	Armor	LoShaelLoOrt	+1 to Lightning Mastery 1% L75 Thunderstorm on kill Level 15-30 Holy Shock Aura 5-15% Lightning Absorb +1-500 Lightning Damage +20% Faster Run/Walk +20% Faster Hit Recovery
Selflessness	Armor	JahJahJahJah	Ethereal Reduce Durability by 99% +450 life Damage Reduced by 60 Magic Damage Reduced by 60 Physical Damage Reduced by 99% Reduce Poison Length 99% 50% lifesteal +20% all max res
The Ghost of Uncertainty	Crossbow	NefLoSurGulDolGul	100% L1 Teleport on hit 25% L20 Decoy on hit 25% L20 Slow Missiles on hit 25% L20 Cloak of Shadows on hit 20% L20 Blade Sentinel on hit 100-500% enhanced damage +3%/lvl enhanced damage

5 Changing basic properties

We add a few recipes to basically transform between:

- Socketed and unsocketed items
- Ethereal and non-ethereal items
- White, magic, rare, unique and set items (not **into** unique and set though).
- Basic, exceptional and elite items.

The recipes are as follows (note the additional costs are sometimes symbolic, sometimes costly)

Basic item conditions	**Additional materials**	**Resulting item**
White, socketable	3 Perfect Skulls, 1 Unique Essence	Socketed version of the same item (random number of sockets)
White, magic or rare; non-ethereal, weapon or armor	2 Eths, 2 Unique Essences	Ethereal version of the same item
White, ethereal	2 Nefs, 2 Unique Essences	Non-ethereal version of the same item (removes other properties)
Weapon or armour	4 Perfect Diamonds	White version of the same item
Can have affixes	3 Perfect Sapphires, 1 Unique Essence	Random magic item of the same type
Can have affixes, not charm	2 Perfect Topazs, 2 Unique Essences	Random rare item of the same type
Weapon or armour; white, magic or rare	4 Sols	Transform base item into exceptional version
Weapon or armour; white, magic or rare	4 Mals	Transform base item into elite version

We also add a recipe to reroll whites, but not using runes as this makes finding perfect whites with skills too easy. Instead, we require 4 flawless gems to reroll a white item. Similarly, we allow 3 flawless gems to be used to repair an item.

6 Augments - Unique abilities for magic and rare items

The last two sections detail the most ground-breaking modifications that the mod includes, that provides both another use for runes/essences and opens up the build possibilities, while also increasing how much the endgame can extend. At least, that is the hope.

The most important type of augments are for wearable items (weapons, armor, rings and amulets), and all of them provide **abilities that do not stack by having them multiple times**. Note that these abilities are **added** to the base item, so finding good magic/rare bases to augment is crucial, making magic/rare item finding even more important, while uniques/runewords

will still be the most useful item in many cases. Similarly, we only allow these to be used on magic/rare items, and even more, we add the rule that only one augment can be applied to rare items, and two augments to magic items. This last condition cannot be imposed easily directly through modding, so instead we add an extra property to all augments (+1% extra experience gained) that is nice, but goes well with the spirit of the mod, and in essence counts how many augments have been applied. The player is responsible for not cheating, and never producing a rare with more than 1% extra experience, or a magic with more than 2% extra experience.

The following table details all of the augments that I have considered. Note that all are applicable to magic and rare, the only difference being whether one or two of them can be applied to the same item. These are similar to runewords, but applied through the cube. The order does not matter, of course.

(UE means Unique Essences and SE means Set Essences)
Note the following legend for which applicable items each code represents:

Code	**Item type**
H	Helms
Am	Amulets
W	Weapons
A	Body armors
S	Off-hands
G	Gloves
R	Rings
Bl	Belts
B	Boots

Effect	Recipe	H	Am	W	A	S	G	R	Bl	B
Sockets (max.)	Cham + Jah + Ber + Sur	X		X	X	X				
Slow enemies by 50%	Ral + Tal					X	X		X	
Piercing attacks	Io + Thul + Eth			X				X	X	
Knockback	Dol + Dol + Ith	X		X			X			
Ignore Defense	Lum + Nef		X		X			X		
Freeze	Ith + Tir + El			X		X		X		
Cannot be frozen	Cham + Lem	X			X					X
Indestruct./Rep. quantity	Zod + Mal + UE	X		X	X	X	X		X	X
Reduce poison dur. 75%	Gul + Um + Um	X				X				X
Prevent monster heal	Fal + Ort		X	X					X	
Enemies rest in peace	Um + Sol		X		X			X		X
Hit causes flee 75%	Gul + 2SE	X							X	X
+3-7 Critical Strike	Jah + Vex + UE			X			X		X	
+2-5 Dodge/Avoid/Evade	Cham + Io + SE	X			X	X				X
20% L15 Decoy when hit +8-12 Decoy	Mal + Mal + UE		X		X			X		
+3-7 Valkyrie	Sur + Pul					X	X	X		
+5-10 Multishot +15-20 Guided arrow +10-15 Strafe	Zod + Lo + 2UE			X		X	X			
+8-12 Burst of Speed	Ber + Sur + 2SE		X		X	X				
+8-12 Fade	Lo + Lo + Gul + Ist		X		X	X				
5% L5 Mind Blast on hit 15% L5 Mind Blast when hit +2-5 Mind Blast	Jah + Ber + Lo + UE	X				X		X		
+3-7 Shadow Master	Ohm + 3UE					X	X	X		
10% L15 Blade Shield when hit +8-12 Blade Shield	Ist + Ist + Ist + UE	X		X	X					
+8-12 Shout +8-12 Battle Orders +8-12 Battle Command	Ohm + Ohm + Um + SE			X		X				
+5-10 Find Item	Ist + Ist + Um + Ko	X			X		X	X		X
+5-10 Double Swing	Lem + Fal + SE			X		X	X			
+5-10 Double Throw	Pul + Ko + SE			X		X	X			
+5-10 Frenzy	Lo + Vex + UE			X		X	X			
+5-10 Whirlwind	Ber + Ohm + SE + UE			X		X	X			
+3-7 Berserk	Zod + Io		X	X		X	X			

Effect	Recipe	H	Am	W	A	S	G	R	Bl	B
20% L15 Cyclone Armor when hit +8-12 Cyclone Armor	Cham + Ohm + Mal				X	X				X
+8-12 Werewolf +8-12 Lycanthropy	Zod + Jah				X			X		
+8-12 Werebear +8-12 Lycanthropy	Ber + Ber + UE				X			X		
+5-10 Feral Rage	Jah + Um + UE			X		X	X			
+5-10 Maul	Jah + Pul + Lem			X		X	X			
+5-10 Fury	Zod + Jah + Jah			X		X	X			
+3-7 Oak Sage	Ber + Gul	X	X			X	X	X		
+3-7 Carrion Vine	Vex + Ist + Lem + Lem	X	X			X	X	X		
+3-7 Heart of Wolverine	Jah + Sur	X	X			X	X	X		
+3-7 Solar Creeper	Gul + Gul + SE	X	X			X	X	X		
+3-7 Spirit of Barbs	Vex + Ko + UE	X	X			X	X	X		
+5-10 Skeleton +5-10 Skeleton Mage +5-10 Skeleton Mastery	Zod + Ber + Ohm + UE					X	X	X		
+2-5 Summon Resist	Ist + Hel + Hel + SE	X	X		X	X	X	X	X	X
+5-10 Revive +5-10 Skeleton Mastery	Zod + Lo + Ist + UE					X	X	X		
+3-7 Clay Golem +3-7 Iron Golem +3-7 Fire Golem +10-15 Golem Mastery	Mal + Um + Shael + Amn					X	X	X		
20% L15 Bone Armor when hit +8-12 Bone Armor	Ber + Gul + Sol				X	X				X
20% L15 Corpse Explosion on kill +3-7 Corpse Explosion	Mal + Um + Pul + UE		X	X		X	X	X		
10% L10 Bone Wall on hit 10% L10 Bone Prison on hit +2-5 Bone Wall +2-5 Bone Prison	Gul + Pul + Lem + Shael	X				X		X		
25% L10 Amplify Damage on hit +3-7 Amplify Damage	Lo + Ohm + Vex			X		X	X		X	
25% L10 Iron Maiden when hit +3-7 Iron Maiden	Ohm + Ist + UE + SE	X			X	X			X	X
25% L10 Life Tap on hit +3-7 Life Tap	Ber + Mal + Mal + Thul			X		X	X		X	
25% L10 Decrepify on hit +3-7 Decrepify	Jah + Lum + Dol + Eld			X		X	X		X	
25% L10 Lower Resist on hit +3-7 Lower Resist	Cham + Ist + Sol + Ral			X		X	X		X	
L10-15 Vigor Aura	Ohm + Gul + Nef + UE	X	X		X	X		X	X	X
L10-15 Meditation Aura	Gul + Um + Io + Io	X	X		X	X		X	X	X
L10-15 Redemption Aura	Vex + Vex + Shael + Shael	X	X		X	X		X	X	X
L10-15 Might Aura	Cham + 3UE		X	X		X				
L10-15 Concentration Aura	Jah + Jah + 2UE		X	X		X				

Effect	Recipe	H	Am	W	A	S	G	R	Bl	B
L10-15 Fanaticism Aura	Zod + Jah + Lo + UE		X	X		X				
L10-15 Thorns Aura	Vex + 2SE + UE	X	X		X	X		X		X
L15-30 Holy Fire Aura	Cham + Ber + UE		X	X		X				
L15-30 Holy Freeze Aura	Zod + Zod + SE + UE		X	X		X				
L15-30 Holy Shock Aura	Cham + Jah + 2SE		X	X		X				
L15-30 Sanctuary Aura	Ohm + Vex + Lem +Hel		X	X		X				
L15-30 Conviction Aura	Jah + Sur + Sur + Vex		X	X		X				
+5-10 Smite	Sur + Vex + Pul			X		X	X			
+5-10 Zeal	Jah + Mal + 2UE			X		X	X			
+5-10 Charge	Lo + Ohm + SE + UE			X		X	X			
+5-10 Vengeance	Sur + Lo + Sol + SE			X		X	X			
+8-12 Holy Shield	Zod + Gul + Pul + Lum		X		X	X				
+8-12 Frozen Armor +8-12 Shiver Armor +8-12 Chilling Armor	Ber + Gul + Um + Um		X		X	X				
15% L30 Frozen Orb on hit +15-25 Frozen Orb	Cham + Cham + Io + Tir			X		X	X		X	
20% L10 Static Field on hit +3-7 Static Field	Zod + Sur + Vex + Mal			X		X	X		X	
+5-10 Telekinesis	Mal + Um + Io + Hel	X			X		X	X		X
+5-10 Teleport	Jah + Jah + Ber + Ber	X			X		X	X		X
+8-12 Energy Shield	Vex + Um + Um + Dol		X		X	X				
+3-7 Warmth	Um + Pul + Lem + Fal			X			X		X	
+20-30 Magic Arrow	Mal + Fal			X			X			
+15-25 Immolation Arrow	Lo + Ohm + UE			X			X			
+15-25 Freezing Arrow	Sur + Sur + SE			X			X			
+5-10 Inner Sight 25% L10 Inner Sight on hit 25% L10 Inner Sight when hit	Um + Lem + Lum			X		X	X		X	
+5-10 Slow Missiles 5% L10 Slow Missiles on hit 10% L10 Slow Missiles when hit	Gul + Pul + Ral			X		X	X		X	
+5-10 Jab	Jah + Um + 2SE			X		X	X			
+15-25 Poison Javelin +15-25 Plague Javelin	Ber + Vex + Thul +Thul			X		X	X			
+5-10 Impale	Ohm + Pul + Pul + Tal			X		X	X			
+15-25 Lightning Bolt +15-25 Lightning Fury	Jah + Jah + Ith + SE			X		X	X			
+15-25 Charged Strike +15-25 Lightning Strike	Zod + Ist + Dol + Eld			X		X	X			

Effect	Recipe	H	Am	W	A	S	G	R	Bl	B
+5-10 Fend	Cham + Gul + Lem + Ko			X		X	X			
+15-25 Inferno	Lo + Gul + Amn +Tir		X	X		X		X		
5% L75 Blaze on kill +15-25 Blaze	Jah + Ohm + Shael + Shael		X	X		X		X		X
30% L30 Fire Ball on hit +15-25 Fire Ball	Ohm + Ist + Ist + SE		X	X		X		X	X	
10% L30 Fire Wall on hit +15-25 Fire Wall	Gul + Mal + Io + Ort		X	X		X		X	X	
1% L30 Enchant on hit +8-12 Enchant	Sur + Pul + Lum + Hel		X	X		X	X	X		
10% L30 Meteor on hit +15-25 Meteor	Zod + Fal + Dol + Nef		X	X		X		X	X	
10% L30 Hydra on hit +15-25 Hydra	Cham + Gul + Shael + Ith	X	X	X		X		X		
30% L30 Charged Bolt on hit +15-25 Charged Bolt	Ist + Ist + Lem + UE		X	X		X		X	X	
50% L30 Nova when hit +15-25 Nova	Vex + Um + Lum + Eth		X		X	X		X		X
15% L30 Lightning on hit +15-25 Lightning 15% L30 Chain Lightning on hit +15-25 Chain Lightning	Jah + Vex + Ko + UE		X	X		X		X	X	
1% L75 Thunder Storm on kill +15-25 Thunder Storm	Zod + Jah + Tir + UE	X	X	X	X	X		X		
50% L30 Frost Nova when hit +15-25 Frost Nova	Gul + Mal + Pul + SE		X		X	X		X		X
30% L30 Glacial Spike on hit +15-25 Glacial Spike	Vex + Um + Pul + Ral		X	X		X		X	X	
10% L30 Blizzard on hit +15-25 Blizzard	Cham + Mal + Ko + Ko		X	X		X		X	X	
30% L30 Teeth on hit +15-25 Teeth	Ohm + Pul + Pul		X	X		X		X	X	
+15-25 Poison Dagger	Ohm + Vex + Vex + El			X		X	X			
20% L30 Poison Explosion on kill +15-25 Poison Explosion	Vex + Mal + Um + Pul		X	X		X	X	X		
30% L30 Bone Spear on hit +15-25 Bone Spear	Jah + Ist + Shael + Nef		X	X		X		X	X	
20% L30 Poison Nova on hit +15-25 Poison Nova	Cham + Vex + Ist + Thul		X	X		X		X	X	

Effect	Recipe	H	Am	W	A	S	G	R	Bl	B
30% L40 Bone Spirit on hit +15-25 Bone Spirit	Zod + Gul + Lem + Sol		X	X		X		X	X	
+5-10 Sacrifice	Pul + Fal + Ko + Amn			X		X	X			
30% L30 Holy Bolt on hit +15-25 Holy Bolt	Um + Um + Lem + SE		X	X		X		X	X	
30% L30 Blessed Hammer on hit +15-25 Blessed Hammer	Lo + Ist + Io + Dol		X	X		X		X	X	
10% L30 Fist of the Heavens on hit +15-25 Fist of the Heavens	Zod + Ber + Sur		X	X		X		X	X	
L10-15 Blessed Aim Aura	Cham + Mal + Um + Eld		X	X		X				
L20-30 Prayer Aura	Ist + Um + Lem + Lem	X	X		X	X		X	X	X
L15-30 Cleansing Aura	Zod + Vex + Tir + El	X	X		X	X		X	X	X
L5-15 Salvation Aura	Sur + Um + Um + Um				X					
+1 Blade Mastery	Mal + Mal + Io + Eth			X		X	X			
+1 Axe Mastery	Um + Pul + Fal + Tal			X		X	X			
+1 Mace Mastery	Ko + Ko + Ko + Lum			X		X	X			
+1 Polearm Mastery	Ist + Lem + Hel + Hel			X		X	X			
+1 Throwing Mastery	Mal + Ko + Dol + UE			X		X	X			
+1 Spear Mastery	Lem + Fal + Fal + Eld			X		X	X			
+1 Increased Speed	Shael + Shael + Shael + Shael				X			X		X
+1 Natural Resistance	Um + Um + Um + Um				X	X		X		
+5-10 Taunt 25% L10 Taunt on hit 25% L10 Taunt when hit	Gul + Eld + 2SE	X			X	X			X	
+5-10 Battle Cry 25% L10 Battle Cry on hit	Ohm + Mal + Dol + Sol	X			X	X			X	
+5-10 War Cry 5% L5 War Cry on hit	Cham + Ist + Lem + Ko	X			X	X			X	
+5-10 Grim Ward 10% L10 Grim Ward on kill	Zod + Ohm + Lum + Thul	X			X	X			X	
+3-7 Plague Vine	Vex + Ist + Lem + Hel	X	X			X	X	X		
+5-10 Spirit Wolf +5-10 Dire Wolf	Cham + Ohm + Pul + Io					X	X	X		
+5-10 Grizzly	Zod + Cham + Mal + Amn					X	X	X		
+15-25 Rabies	Ohm + Vex + Vex + El			X		X	X			
+15-25 Fire Claws	Jah + Pul + Sol + Sol			X		X	X			
30% L30 Firestorm on hit +15-25 Firestorm	Jah + Um + Hel + Dol		X	X		X		X	X	
15% L30 Molten Boulder on hit +15-25 Molten Boulder	Jah + Ohm + Fal + Eth		X	X		X		X	X	

Effect	Recipe	H	Am	W	A	S	G	R	Bl	B
+15-25 Arctic Blast	Cham + Um + Um + Eld		X	X		X		X		
10% L30 Fissure on hit +15-25 Fissure	Zod + Jah + Gul + Ist		X	X		X		X	X	
15% L30 Twister on hit +15-25 Twister 15% L30 Tornado on hit +15-25 Tornado	Lo + Mal + Um + Ith		X	X		X		X	X	
5% L30 Volcano on hit +15-25 Volcano	Zod + Ber + Ber + Tir		X	X		X		X	X	
1% L50 Armageddon on kill +15-25 Armageddon	Jah + Sur + Sur + Tal		X	X		X		X		X
1% L50 Hurricane on kill +15-25 Hurricane	Cham + Sur + Lo + Thul		X	X		X		X		X
30% L30 Fire Blast on hit +15-25 Fire Blast	Sur + Lo + Ist + Ith		X	X		X		X	X	
10% L30 Shock Field on hit +15-25 Shock Field	Ber + Mal + Um + UE		X	X		X		X	X	
+15-25 Charged Bolt Sentry +15-25 Lightning Sentry +15-25 Death Sentry	Lo + Ohm + Ist + SE		X	X		X		X	X	
+15-25 Wake of Fire +15-25 Inferno Sentry	Lo + Lo + Um + Nef		X	X		X		X	X	
30% L15 Blade Sentinel on hit +15-25 Blade Sentinel	Jah + Gul + Lem + Io		X	X		X		X	X	
+8-12 Blade Fury	Zod + Lo + Pul + Pul		X	X		X	X	X	X	
+5-10 Cloak of Shadows 5% L10 Cloak of Shadows on hit 10% L10 Cloak of Shadows when hit	Ist + Ko + Dol				X	X			X	X
1% L20 Venom on hit +8-12 Venom	Zod + Vex + Dol + Dol		X	X		X	X	X		
+15-25 Dragon Talon	Ohm + Gul + Fal + Ral		X	X		X		X		X
50% L30 Ring of Fire when hit +15-25 Ring of Fire	Jah + Mal + Io + Ral		X		X	X		X		X
10% L30 Flame Wave on hit +15-25 Flame Wave	Ber + Sur + Lum + Eth		X	X		X		X	X	
20% L30 Apocalypse on kill +15-25 Apocalypse	Zod + Zod + 2 UE		X	X		X	X	X		
25% L10 Sigil:Rancor on hit +3-7 Sigil:Rancor	Gul + Mal + Lem			X		X	X		X	

Effect	Recipe	H	Am	W	A	S	G	R	Bl	B
20% L10 Sigil:Death on kill +3-7 Sigil:Death	Vex + Ist + Pul + Eld		X	X		X	X	X		
30% L30 Miasma Chain on hit +15-25 Miasma Chain	Lo + Um + Thul + SE		X	X		X		X	X	
10% L30 Abyss on hit +15-25 Abyss	Zod + Mal + Ko + Dol		X	X		X		X	X	
+5-10 Echoing Strike	Jah + Ist + Dol + Ith			X		X	X			
+5-10 Cleave	Gul + Gul + Lem + SE			X		X	X			
20% L15 Psychic Ward when hit +8-12 Psychic Ward	Cham + Gul + Fal + Fal				X	X				X
1% L60 Eldritch Blast on kill +15-25 Eldritch Blast	Jah + Sur + Pul + Tir	X	X	X	X	X		X		
+5-10 Mirrored Blades	Ber + Ist + Tal + Ith			X		X	X			
+15-25 Hex: Purge	Lo + Mal + Hel + Sol			X		X	X			
+5-10 Summon Goatman +1 Demonic Mastery +1 Blood Oath +1 Death Mark	Sur + Sur + Ist + Lem					X	X	X		
+5-10 Summon Tainted +1 Demonic Mastery +1 Blood Oath +1 Death Mark	Lo + Ohm + Vex + Thul					X	X	X		
+5-10 Summon Defiler +1 Demonic Mastery +1 Blood Oath +1 Death Mark	Ber + Um + Um + Ort					X	X	X		
+5-10 Bind Demon +1 Demonic Mastery +1 Blood Oath +1 Death Mark	Zod + Cham + Cham + Cham					X	X	X		
+8-12 Consume	Jah + Nef + Tir + El		X		X	X				
+15-25 Blood Boil +15-25 Engorge	Vex + Gul + Ist + Ral		X	X		X		X	X	

Effect	Recipe	H	Am	W	A	S	G	R	Bl	B
10-20% Reanimate as Flesh Hunter	Ist + Ist + Ist + Um	X			X	X	X	X		X
8-15% Reanimate as Serpent Magus	Ohm + Um + 2UE	X			X	X	X	X		X
3-8% Reanimate as Mauler	Ber + Gul + Ko + Eth	X			X	X	X	X		X
6-12% Reanimate as Undead Soul Killer	Jah + Mal + Mal + Ith	X			X	X	X	X		X
1% Reanimate as Duriel	Zod + Zod + Zod + Zod	X			X	X	X	X		X
2-4% Reanimate as Clay Golem 2-4% Reanimate as Blood Golem 2-4% Reanimate as Iron Golem 2-4% Reanimate as Fire Golem	Lo + Vex + Ist + Um	X			X	X	X	X		X
3-8% Reanimate as Bear	Ber + Ber +Tir + Tir	X			X	X	X	X		X
3-10% Reanimate as Hell Lord	Jah + Vex + Nef + El	X			X	X	X	X		X

7 Augments - Long-term progression

Finally, the purpose of the augments specified here is to increase how far long-term progression can go, giving a more relevant use to charms and jewels/sockets, without giving too strong effects that completely overshadow everything else. Note that we consider charm/jewel augments to only be applicable once, even to magical items. The properties added are stackable ones that benefit from multiple instances but which are not commonly present in magic/rare items. The numbers are also small in each case so that even when stacking they do not amount to too high values.

Effect	Recipe	SC	LC	GC	JEW
Crushing blow	Jah + 2UE	3%	5%	7%	7%
Damage reduced	Ber + SE	2%	3%	5%	5%
Max all resists	Ber + Lo + Ohm			3%	3%
All absorbs %	Cham + Sur + Sur			1%	
Max. life %	Vex + Vex + Dol	2%	3%	5%	5%
Max. mana %	Ist + Mal + Io + Io	4%	6%	10%	10%
Regenerate mana	Gul + Um + Um + Lem	3%	5%	6%	6%
All skills	Zod + Ohm			1	1
FHR	Sur + SE + UE				20%
FCR	Sur + Lo + 2SE	3%	5%	7%	7%
IAS	Jah + Lem + Io	5%	8%	10%	10%
FRW	Cham + Hel	2%	3%	5%	5%
Damage Goes to Mana	Ist + Pul + Lum	10%	20%	30%	30%
Magic Find	Ist + Ist + 2UE	5%	10%	15%	15%
Gold Find	Ist + Um + 2SE	25%	50%	75%	75%
Reduce vendor prices	Vex + Gul + Ist + UE			2%	2%
Mana after kill	Mal + Pul + Fal + Sol	3	5	8	8
Life after kill	Mal + Um + Um + UE	2	3	5	5
Fire skill damage	Jah + Jah + UE	1%	2%	3%	5%
Cold skill damage	Cham + Jah + UE	1%	2%	3%	5%
Lightning skill damage	Zod + Jah + UE	1%	2%	3%	5%
Poison skill damage	Jah + Vex + Vex + UE	1%	2%	3%	5%
Magic skill damage	Sur + Ohm + Ohm	1%	2%	3%	5%
Reduce fire resistance	Zod + Jah + Jah	1%	2%	3%	5%
Reduce cold resistance	Zod + Cham + Jah	1%	2%	3%	5%
Reduce lightning resistance	Zod + Zod + Jah	1%	2%	3%	5%
Reduce poison resistance	Zod + Jah + Vex + Vex	1%	2%	3%	5%
Reduce magic resistance	Ohm + Ohm + Gul + Gul	1%	2%	3%	5%
Reduce physical resistance	Zod + Cham + Ohm + Um	1%	2%	3%	5%
Damage % increased per level	Cham + Jah + Gul + Gul	0.125	0.25	0.375	0.375
Bonus to attack rating per level	Ber + Lo + Lum + Ith	0.5	1	1.5	1.5

8 Getting gems

One issue that arose when using the systems mentioned above is the consequence of the following factors:

- We made high runes easier to obtain by using perfect gems to produce them, in high quantities.

- We made high runes usable for many more things via augments.
- We made runes downconvertible by rerolling magic/rare items.
- We also made good magic/rare items easier to obtain through this.

But a key factor remains:

- The only way to obtain perfect gems is still the regular ways, and the amounts required to produce enough high runes to utilize both rune words AND augments is huge.

Originally, I thought that finding suitable magic/rare items would be a lot easier than getting the runes and thus used runes to downconvert them into gems. However, while this worked, it made runes have two primary uses; specially mid to low level ones. This meant that farming Countess became pretty much a must.

Instead, we leave the reroll functionality of runes, making mid to low runes very valuable, we keep a recipe that converts an El rune into a perfect gem, to give them some extra value and contribute to the gems; and we introduce a new way to obtain perfect gems.

We will use magic and rare rings and amulets. These are small, meaning they can be picked up and stored without it being a big hassle and it's specific enough that we are not picking up every item or that we are flooded in gems. There is also no specific source of large amounts of these, and they are not really useful in large quantities for anything else. We diversify the resources.

Thus, we add two recipes:

- Magic ring $\rightarrow$ 3 perfect gems
- Magic amulet $\rightarrow$ 2 magic rings of the same item level.
- Rare ring $\rightarrow$ 2 magic rings of the same item level.
- Rare amulet $\rightarrow$ 1 rare ring and 1 magic amulet of the same item level.

Thus, when we need to reroll magic or rare items, Countess might be a good farming target. When we need gems for upgrading runes, we can do something that drops large amounts of magic items (like, for example, Cows). Having rare items give more also means having Magic Find represents an advantage.

9 Keeping high level areas relevant

A lot of the above changes worked really well for quite a while, with only one very minor issue: High level areas became considerably less relevant. The game more or less became effectively about 4 main things: Finding gems (rings/amus), finding unique/set essences, finding low level runes to reroll items, and finding good blue/yellow bases to reroll. This is a decent balance, but finding bases can be done in many different places, like for example Cows. Cows is clearly one of the best places for rings/amus as well. Mephisto with high magic find is incredibly superior for unique/set essences. Countess can provide lots of low level runes. This is a decent variety but it's missing a bigger incentives for high level areas like lvl 85 areas. Adding uniques with level 94 helped with this, but we wanted something more solid.

The main distinguishing factor of these areas is the combination of being able to drop TC85 and high monster level that allows them to drop the highest level uniques/set items in the game. Without any distinguishing feature to make these more relevant than any other elite unique/set items, this isn't an incentive. Some of these items are still really good to use directly and so there is some incentive, but it's not enough by itself.

As a consequence, we decided to add two small, secondary incentives that do not modify the core balance above, but add another extra nice layer that can only be achieved through these.

The first change was to make all TC85 blue or rarer items have some value. As such, two additional recipes were added:

- Any magic item from TC85 → Random organ
- Any rare item from TC85 → Two random organs

There are 7 types of organs: Hearts, Brains, Eyes, Spleens, Jawbones, Fangs, and Horns. Another source of random organs are 12 flawless gems, which give 1 organ, 1 random jewelry, and 1 perfect gem. This was added as converting flawless gems into perfect gems became irrelevant when perfect gems could be obtained from rings and amulets in much larger numbers. To a large degree, flawless gems are now more valuable than perfect gems, but this makes a lot of sense given how they are obtained now. There are two families of cube recipes that organs can be used for. One generates random valuable items (3 organs of the same type result in 1 rune, 1 random top jewelry, and 1 random ring), the other creates special rare belts that can be

potentially really good, but with low probabilities for good combinations of statistics. We chose belts because they are overall the least exciting piece of equipment with not really that many great options, even with Augments.

Inputs	Output
Eye, brain, heart, spleen, fang	Random rare belt, with potentially additional properties... +1-3 All Skills (5% chance) 10-40% Faster Cast Rate (15% chance) Increase Maximum Mana 10-25% (25% chance)
Jaw, horn, fang, spleen, heart	Random rare belt, with potentially additional properties... +25-100 Maximum Damage (5% chance) +10-40% Increased Attack Speed (15% chance) +5-15% Bonus to Attack Rating (25% chance)

Much more important is the second change. We picked some of the rarest, most iconic high level sets and uniques, and added recipes that would allow to imbue some of their most iconic properties to grand charms. This does not count as an augment and so these charms can still receive one regular augment on top of this, but the same charm cannot be imbued like this more than once. We specifically chose set items and unique items that are very rare and/or very iconic, and specifically, none of these items, except for amulets and rings (and Arachnid Mesh, which is a bit special), can be dropped by Mephisto or lower TC/MLVL monsters. This doesn't force to only farm lvl 85 areas, but it means these items are far more common in higher level areas. Rarer items also receive more powerful bonuses as a rule, and the bonuses stay in the spirit of what builds the item would normally benefit.

Note that because it's not possible, from the cube recipe, to distinguish different set/unique items with the same base, we either allow both or, in the case of amulets and rings, we use additional keys in the recipe to identify them. This allows cheating to some degree but the player will just not cheat.

Item	Effects
Arcanna's sign (amulet) + 1 key	+100% magic find
Aldur's deception (chest)	+2 druid +15 all stats
Cow King's horns (helm)	+500% gold find
Telling of beads (amulet) + 2 keys	+1 all skills +10 all res + 20 all stats
Griswold's redemption (scepter)	+2 pala +3% all max res
Immortal King's soul cage (chest)	+2 barb +50 strength
M'avina's true sight (diadem)	+2 ama +20% run/walk
Natalya's mark (claws)	+2 assa +50 life
Tal Rasha's adjudication (amulet) + 3 keys	+2 sorc + 20% FCR
Trang-Oul's guise (helm)	+2 nec +100 energy
Horazon's Legacy (boots)	+2 war +50% mana regen
Stone of Jordan + 1 key	+1 all skills +50 mana
Bul-Kathos wedding band + 2 keys	+1 all skills +50 life
Wisp projector + 3 keys	5% L50 Chain Lightning on hit 2% L50 Fist of the Heavens on hit 5% Reanimate as Burning Soul
Seraph's hymn + 1 key	+1 all skills +200 AR
Metalgrid + 2 keys	+35 all res +350 defense
Griffon's eye	+10% lightning damage -10% lightning resistance
Darkforce spawn	+2 nec+20% FCR
Halaberd's reign	+3 shouts
Death's fathom	+25% cold damage
Ravenlore	+2 elemental skills (druid)
Dragonscale	+10% fire skill damage +300-500 fire damage
Crown of ages	5% reduced damage +10 all res +15% FHR
Leviathan	15% reduced damage
Steel carapace	+1000 defense
Tyrael's might or Templar's might	+20% FRW +20 all res +300 defense Cannot be frozen
Steelrend	+40% ED
Shadow dancer	+10% FRW +2 shadow disciplines
Arachnid mesh	+1 all skills +10% FCR
Deathcleaver	+25% ED +20% IAS
Messerchdmidt's reaver	+0.75/lvl enhanced max damage %
Executioner's justice	+75% ED
Windforce	+30% ED +0.5/lvl max damage
Earth shifter or Cranium Basher	+4 elemental skills (druid)
Astreon's iron ward	+3 combat skills (pala)
Mang Song's lesson	+3 all skills -5% fire/cold/lightning resis
Grandfather	+30% ED +0.5/lvl max damage

Item	Effects
Death's web	-30% enemy poison res
Sazabi's cobalt redeemer	+100% damage to demons
Fleshripper	25% open wounds Prevent monster heal
Schaefer's hammer or Stone crusher	5% chance to cast level 10 static field on strike
Tomb reaver	+20% IAS +10 all res +20% magic find
Boneflame	+2 nec +10% FRW +10 all res
Ars Al'Diabolos	+7 apocalypse
Ars Dul'Mephistos	-15% enemy magic res
Ars Tor'Baalos	+3 demon skills

10 Keys and Essences for single game bonuses

Another two types of item that are supposed to be valuable, and encourages farming variety, but which can become pointless depending on the situation, are keys for the pandemonium event and essences used for reset tokens. These are particularly useless when there is a big imbalance between ones and others for other reasons (for example, excess of Keys of Terror due to Countess runs).

Similarly, one underexploited feature of Diablo 2 involves temporary bonuses from consumable items. With scarce items, and very powerful bonuses that last about half an hour, this can help farm very high level areas while encouraging making the most of the run (since the bonuses are lost when quitting). We introduced recipes to generate random special potions from 12 essences (plus a random top jewelry and a random amulet), and an organ, a rune, and a random top jewelry from 6 keys.

Special potion	Effects
Skill Potion	+15 All Skills for 30 minutes
Elemental Potion	+500% fire, cold, and lightning damage for 30 minutes
Poison Potion	+500% poison damage for 30 minutes
Damage Potion	+5000 damage for 30 minutes
Survival Potion	+5000 Life and Replenish Life +2500 for 30 minutes

11 Jewelry and value

At this point, basically it is possible to only pick up items that will be kept one way or another for some purpose. This is good. Especially with the

ability to reroll magic and rare items, which made it so we only needed to care about the base rather than the actual mods. Most magic and rare items can be gambled too if needing a good base. With the exception of charms and jewels. One option was to allow gambling of charms and jewels, but a more interesting option was to give charms and jewel additional value, so that once picked up, even if bad rolls and not good for rerolling due to item level, they still had some use. Moreover, this kept the value of high item level charms for rerolling, especially over 90, which are hard to get and make for an additional reason to farm high end bosses.

First, we define the generic Jewelry type to include the following:

- Unique and set essences.
- Rings and amulets.
- Charms.
- Jewels.
- Runes.
- Flawless and Perfect gems.
- Organs as defined above.
- Scroll of knowledge (explained below).
- Keys of Terror, Hate, Destruction.
- Diablo's Horn, Baal's Eye, and Mephisto's Brain (as used for chaotic Tristram).
- Essences of Suffering, Hatred, Terror, Destruction.
- Token of Absolution.
- Special potions.
- Standard of Heroes.

Then, we add cube recipes that either 3 small charms, or 3 jewels will result in a random Jewelry. This can sometimes be really good, with a high level rune, an organ or a scroll of knowledge; or quite bad, with a perfect gem, a bad magic ring, or a low rune. One grand charm produces one random

misc in general, which can be better but can also be worse, this makes it so that holding grand charms is not a thing.

We additionally define Top Jewelry to be specifically concentrated unique and set essences, organs, scrolls of knowledge, Keys and chaotic Tristram organs, special potions, and well as Standards of Heroes. We then add a recipe that a Standard of Heroes will result in 1 Top Jewelry, a set ring, and a random jewelry, giving more incentive for chaotic Tristram.

A scroll of knowledge is a new item that can only be obtained through these recipes that is used to make unique and set items more attractive, by adding a recipe that allows adding 2 sockets to a unique or set item with a scroll of knowledge, or give increased stack size and replenish quantity to throwable weapons.

We also define Basic Jewelry to be rings, amulets, charms, jewels, runes, flawless and perfect gems. We add a 25% chance for unique and super uniques in terror zones only to drop one piece of basic jewelry on top of their other drops, to incentivize non-bosses, and a 1% chance to drop top jewelry for uniques only, to incentivize explorative play.

For Large Charms, which are normally not very useful, we do something a little different. We make a recipe that makes a new random Large Charm from 3 Large Charms, with potential added small bonuses, making these potentially better than small or grand charms, but a lot harder to obtain. Each individual additional attribute has a 20% chance to appear at all. The potential additional attributes are:

- +1-10 Vitality
- +1-5 All Resistances
- +1-50% Gold Find
- 1-10% Faster Hit Recovery
- 1-10% Faster Run/Walk

12 Gold

We also wanted to make both some additional ways to make gold easier to obtain in large quantities and in general incentivize killing large amounts of

low level enemies, rather than focusing exclusively on elites. In particular, we make regular keys have quite a bit of gold value, so they can be sold to NPCs for gold and are worth picking up. We also increase the maximum stack size for keys to 100. Each key is worth 2500 gold

While we want gambling to still be one of the main ways to use gold, both to obtain good bases and rings/amulets to generate perfect gems for upgrading runes, sometimes there just is not a good target to gamble for, since we are not looking for any particular item or base that we do not have like that, nor for additional gems to upgrade runes. Thus, we add a new relatively inefficient item to buy at NPCs, a Raw Jewel, costing 150000 gold and resulting in a random piece of Jewelry in the horadric cube. This, again, could be really good if landing a really valuable Jewelry like a high l evel rune, Standard of Heroes, or organ; or a waste of money, if landing a perfect gem or bad magic ring.

13 Consuming low gems

Temporary consumables add a layer of benefit to playing in the same game for a while, and it can be a way to give use to certain items without significantly increasing clutter. We do this with low level gems that are usually unused. These bonuses last for 5 minutes.

Gem type	**Bonus**	**Amounts (chipped, flawed, regular)**
Sapphire	Regenerate Mana	10%, 25%, 50%
Ruby	Replenish Life	10, 25, 50
Amethyst	Run/Walk Speed	10%, 20%, 30%
Emerald	Increase Experience Gained	10%, 25%, 50%
Topaz	Magic Find	25%, 50%, 100%
Diamond	Gold Find	25%, 50%, 100%
Skull	Physical Damage Reduced	5%, 10%, 15%

14 Augmented Torches

To add yet another nice layer of soft long-term progression and build variability and give more use to set amulets and rings we create recipes to augment hellfire torches.

We add recipes to enhance torches with small and very limited random changes. In particular, we enable them to give small bonuses to class skills. The recipes take as input a torch and a number of keys that identifies the class we want to augment it for:

Class	Number of keys
Amazon	0
Sorceress	1
Necromancer	2
Paladin	3
Barbarian	4
Druid	5
Assassin	6

To limit how many times it can be performed, we add additional final augments. These are in the flavour of hellfire torch and use the reanimate ability to add interesting but not game breaking aspects. Each time the torch is augmented, there is a probability each of these final augments are added. It is technically possible for multiple of these to be added at once, which would make extra lucky torches. Conceptually, a torch with one of these cannot be augmented further. Each attempt gives a random class skill a +2 bonus, and there is overall approximately a 22% chance that at least one of the final augments will be added on each augment.

Final augment	Chance to be added
10% chance to reanimate as Succubus	10%
10% chance to reanimate as Balrog	7%
5% chance to reanimate as Minion of Destruction	5%
1% chance to reanimate as Izual	3%

Statistically, this means that there will be on average 4-5 augments before a final augment is added. In other words, in the long term, the bonus is useful, but not game breaking, and it has a potential to be really good, but is very unlikely to reach that level. This also might incentivize rolling new torches and getting new torches, which keeps the pandemonium event permanently relevant.

Each augment requires 2 set amulets or 3 set rings.

We also add a recipe that transforms a torch into 2 set amulets and a random top jewelry, for excess torches for the wrong classes.

15 Terrorized zones

The basic modification is that by default the level is 7 levels above character level, for a maximum of 106 (109 on uniques). We also add jewelry drops to uniques and super uniques in them, as explained above.

We also group terror zones together from different acts for larger pools for more effective herald hunting, and add some especial tweaks to treasure classes in them.

Finally, we also make monsters above level 100 scale polynomially in health, damage, and experience, and make terror zones get higher and higher levels as going down an act.

15.1 The Cow Level

This stays by itself but we add some extra jewelry drops from them for efficient farming. More frequent heralds but obviously just one area gets relatively restricted.

- Additional chance of uniques.
- Additional top jewel and top rune drops.
- More frequent heralds.

15.2 The Countess

The tower stays by itself but we add some extra rune drops from countess.

- Additional rune drops.

15.3 Travincal

Stays by itself but we increase the chance of uniques. No heralds.

- Additional chance of uniques.
- No heralds.

15.4 Chaos Sanctuary

Stays by itself but we make Diablo have really good chance of uniques. More frequent heralds but obviously just one area gets relatively restricted. Significantly increased XP.

- Additional chance of uniques.
- More frequent heralds.
- Significantly increased XP.

15.5 Worldstone Keep and Throne of Destruction

Stays by itself but we make Baal have really good chance of uniques. No heralds.

- Additional chance of uniques.
- No heralds.

15.6 Nihlathak and Summoner

We effectively double the chance of pandemonium keys when terrorized.

- Double chance of keys.

15.7 The Smith

The Smith has a very weird treasure class normally, with guaranteed drops and being able to drop whites. We turn terrorized Smith into a White pinnata. No additional chances to drop magic, rare, sets, or uniques, but has the best treasure class and always 6 drops.

- Always 6 items.
- Mostly whites.
- Low chance of organs.

15.8 Blood Raven

Like the smith, we make this mini-boss something special. In this case, we make her always drop exactly 1 piece of jewelry, with relatively standard rarity boosts. Blood Raven always gives a herald token.

- Always 1 piece of jewelry (in the conventional sense - charms, rings, amulets, jewels).
- Low chance of organs.
- Guaranteed herald token.

15.9 Izual

We make this mini-boss something special. In this case, we make him always drop exactly 1 set item.

- Always 1 set item.
- Low chance of organs.

15.10 Hephasto

We make this mini-boss something special. In this case, we make him always drop exactly 1 unique item.

- Always 1 unique item.
- Low chance of organs.

15.11 Maggot Lair

This map sucks. We give it super boosted XP.

15.12 Durance of Hate Level 3

Heralds cannot spawn here, but each elite here generates a token for certain.

15.13 Kurast temples

Each of this temples has 50% chance to generate heralds and 50% chance to spawn them, while the Kurast outside regions cannot spawn them.

15.14 Accumulated dungeons

In some caves and dungeons, we make earlier levels generate tokens more frequently but unable to spawn, with the last, smaller level forcing them to spawn. This creates one big splash in the final level that also adds some fun to things.

- Glacial Trail → Drifter Cavern
- Cave Level 1 → Cave Level 2
- Ancients' Way → Icy Cellar
- Undeground Passage 1 → Underground Passage 2
- Pit Level 1 → Pit Level 2

15.15 Extra XP

Some areas give extra XP because they're not that great overall.

- Great Marsh

15.16 Doll dungeons

The swampy pit and flayer dungeon are quite unlikeable, so we add extra XP to make them worth it.

15.17 Cross-act groups

The following groups spawn together.

- Burial Grounds + Great Marsh
- Dry Hills + Outer Steppes + Glacial Trail
- Catacombs + Far Oasis
- Spider Forest + River of Flame + Crystalline Passage
- Blood Moor + Cold Plains + Ancient's Way
- Tal Rasha's Tomb + Durance of Hate
- Dark Wood + Kurast

- A2 Sewers + Nihlathak Halls
- Jail + Lost City
- Flayer Jungle + Bloody Foothills
- Tristram + Monastery
- Rocky Waste + Arreat Plateau + Frozen Tundra

16 Over level 100 uniques

These are exclusive to terror zones. There are two types here. The main one is *enhanced uniques* - versions of normal uniques with some additional nice to have properties. These are not enough to make or break their balance or make them required, but they give something nice to look forward to in these terror zones. Some of them add little bonuses to already useful uniques, and some try to make other uniques a little more useful by adding uncommon but not game changing properties, in some cases they add just what is necessary to make a useless unique become useful.

16.1 Enhanced Uniques

They all get 1% extra experience gained and drop half as frequently as the regular item (so one out of every 3 in these areas will be enhanced), with some exceptions.

Name	Base item	Lvl	Extra Properties
Bloodfist	Heavy Gloves	100	+100 Attack Rating
Magefist	Light Gauntlets	105	+20% Faster Hit Recovery
The Stone of Jordan	Ring	115	+10% Faster Cast Rate
Arm of King Leoric	Tomb Wand	107	+1 All Skills
Blackbog's Sharp	Cinquedeas	107	+5% poison skill damage
Ribcracker	Quarterstaff	105	+5-10% mana leech +25% bonus to attack rating
Buriza-Do Kyanon	Ballista	118	Level 10 Fanatism aura
Valkyrie Wing	Winged Helm	105	+10-20 All Resistances +10-15% Reduced Damage
Vampire Gaze	Grim Helm	110	+10-20 All Resistances +15% Crushing Blow
Skin of the Vipermagi	Serpentskin Armor	112	+20% Faster Hit Recovery
Que-Hegan's Wisdom	Mage Plate	102	+10-25% maximum life
Lidless Wall	Grim Shield	100	+100-150% Gold Find
Waterwalk	Sharkskin Boots	100	+30% Faster Hit Recovery
War Traveler	Battle Boots	107	+30% Faster Hit Recovery
Gore Rider	War Boots	107	+20% bonus to attack rating
String of Ears	Demonhide Sash	105	+20% Increased Attack Speed
Snowclash	Battle Belt	107	+1 All Skills
Thundergod's Vigor	War Belt	110	+5% lightning skill damage
Harlequin Crest	Shako	112	+5-10 Find Item
Arkaine's Valor	Balrog Skin	107	Level 10 Vigor aura
Stormshield	Monarch	105	Level 5 Holy Freeze aura
Messerchdmidt's Reaver	Champion Axe	102	Slain Monsters Rest in Peace +50-100 Damage
The Cranium Basher	Thunder Maul	107	Level 10 Concentration aura
The Grandfather	Colossus Blade	121	25% Crushing Blow
Windforce	Hydra Bow	121	25% L10 Decrepify on hit
Bul-Kathos' Wedding Band	Ring	110	+10% Increased Attack Speed
Mara's Kaleidoscope	Amulet	110	+10% Faster Cast Rate
Raven Frost	Ring	105	+5-10 All Resistances

Arreat's Face	Slayer Guard	107	+15-30% Increased Attack Speed
Homunculus	Hierophant Trophy	102	+2 to Summoning Skills
Titan's Revenge	Ceremonial Javelin	112	+20% Increased Attack Speed
The Oculus	Swirling Crystal	110	+50-100% Gold Find
Herald of Zakarum	Gilded Shield	112	+5 All Max Resistances
Bartuc's Cut-Throat	Greater Talons	107	+1.5/lvl maximum damage
Jalal's Mane	Totemic Mask	110	+10% life leech +15% crushing blow
Tomb Reaver	Cryptic Axe	110	10% L10 Burst of Speed on hit
Death's Web	Unearthed Wand	121	+5-15% poison skill damage
Azurewrath	Phase Blade	110	10% L20 Fist of the heavens on hit
Shadow Dancer	Myrmidon Greaves	110	+1-3 Dragon Talon +1-3 Dragon Tail +1-3 Dragon Flight
Cerebus' Bite	Blood Spirit	107	10% L15 Venom on hit
Tyrael's Might	Sacred Armor	125	Reduce Poison Length 95% Level 10 Concentration aura
Runemaster	Ettin Axe	110	+25-75 maximum damage
Deathcleaver	Berserker Axe	112	+10% Crushing Blow
Executioner's Justice	Glorious Axe	110	+50% bonus to attack rating
Leviathan	Kraken Shell	112	+20 Vitality
Mang Song's Lesson	Archon Staff	125	Level 15 Conviction Aura
The Reaper's Toll	Thresher	107	+30% Increased Attack Speed
Alma Negra	Sacred Rondache	110	+5 to Mace Mastery
Darkforce Spawn	Bloodlord Skull	112	+1 to Necromancer skills
Blood Raven's Charge	Matriarchal Bow	107	+500-1000 magic damage
Shadowkiller	Battle Cestus	102	+1.25/lvl% Enhanced Damage
Gimmershred	Flying Axe	100	5% L25 Meteor on hit 5% L25 Chain Lightning on hit 5% L25 Frozen Orb on hit
Warshrike	Winged Knife	100	+25-75 maximum damage
Griffon's Eye	Diadem	125	+3-7 to Lower Resistances
Thunderstroke	Matriarchal Javelin	115	+15% lightning skill damage
Steelpillar	War Pike	102	+20-45% Increased Attack Speed
Nightwing's Veil	Spired Helm	112	+25% Faster Cast Rate
Crown of Ages	Corona	118	Level 5 Holy Freeze aura
Ravenlore	Sky Spirit	115	+5% fire skill damage
Boneshade	Lich Wand	107	+1-2 to Poison & Bone skills

Death's Fathom	Dimensional Shard	125	+1-2 to Cold Skills
Wolfhowl	Fury Visor	112	Level 10 Fanatism aura
Ormus' Robes	Dusk Shroud	110	+10-25% maximum life
Stormlash	Scourge	118	15% L15 Nova on hit
Dracul's Grasp	Vampirebone Gloves	112	10% Increased Attack Speed
Eschuta's Temper	Eldritch Orb	115	-3-5% fire resistance -3-5% lightning resistance
Sandstorm Trek	Scarabshell Boots	105	+15-30% Magic Find
Heaven's Light	Mighty Scepter	107	+25% bonus to attack rating
Arachnid Mesh	Spiderweb Sash	121	+20% Faster Hit Recovery
Verdungo's Hearty Cord	Mithril Coil	121	+10% Increased Attack Speed
Astreon's Iron Ward	Caduceus	118	2-5 sockets
Nord's Tenderizer	Truncheon	105	-10-20% cold resistance
Earth Shifter	Thunder Maul	125	+25-50% fire skill damage
Steelrend	Ogre Gauntlets	112	+10-25% bonus to attack rating
Seraph's Hymn	Amulet	102	Level 20-30 Sanctuary Aura
Lycander's Flank	Ceremonial Pike	112	6 sockets
Lycander's Aim	Ceremonial Bow	112	5 sockets
Jade Talon	Wrist Sword	102	+100% poison skill damage
Firelizard's Talons	Feral Claws	107	20% L30 Enchant on hit +10-25% fire skill damage
Boneflame	Succubus Skull	115	+1 Teleport
Dragonscale	Zakarum Shield	107	Level 50 Holy Fire Aura 2-4 sockets
Twitchthroe	Studded Leather	100	10-20% reduced damage 4 sockets
Chance Guards	Chain Gloves	105	+2 to random class skills
Frostburn	Gauntlets	110	+20-40% faster hit recovery +20-50% faster cast rate
Nightsmoke	Belt	100	50% L50 Energy Shield when struck
Goldwrap	Heavy Belt	107	+1 All Skills
Wizendraw	Long Battle Bow	115	50% L30 Blizzard on hit 6 sockets
Ume's Lament	Grim Wand	110	+7 to random necromancer skill
Rockstopper	Sallet	102	L20-30 Thorns Aura
Stealskull	Casque	107	L10-20 Blessed Aim Aura

Duriel's Shell	Cuirass	105	20% L15 Bone Armor when hit 20% L15 Cyclone Armor when hit
Skullder's Ire	Russet Armor	118	+2 to random class skills
Guardian Angel	Templar Coat	118	+1-3 Combat Skills (paladin) +1-3 Offensive Auras (paladin) +1-3 Defensive Auras (paladin)
Lava Gout	Battle Gauntlets	110	+15-45% fire skill damage
Infernostride	Demonhide Boots	102	100% L50 Blaze on kill +10-20% fire skill damage
Gloom's Trap	Mesh Belt	112	L40-50 Meditation Aura
Islestrike	Twin Axe	110	10% L20 Heart of Wolverine on hit
Spellsteel	Bearded Axe	118	+1-7 All Skills +7 to random class skills
The Minotaur	Ancient Axe	100	+1-6 sockets
Riphook	Razor Bow	102	Knockback 25% L15 Amplify Damage on hit
Kuko Shakaku	Cedar Bow	112	L50 Holy Fire Aura
Goldstrike Arch	Gothic Bow	110	L50 Sanctuary Aura (modify)15% L25 Fist of Heavens on hit
Pus Spitter	Siege Crossbow	105	(modify)10% L30 Poison Nova on hit (modify) 15% L10 Lower Resist on hit 100% L30 Poison Explosion on kill +10-20 to Venom
Fleshrender	Barbed Club	100	+1.25/lvl% enhanced damage
Athena's Wrath	Battle Scythe	100	+1-5 sockets
Pierre Tombale Couant	Partizan	100	+1-5 sockets
Husoldal Evo	Bec-De-Corbin	102	L10-20 Might Aura
Hone Sundan	Yari	105	40% increased attack speed 50% bonus to attack rating
Spire of Honor	Lance	105	Ethereal Indestructible 1-6 sockets
Chromatic Ire	Cedar Staff	110	(modify)+3-5 to All Skills (modify)+1-7 to Cold Mastery (any class) (modify)+1-7 to Lightning Mastery (any class) (modify)+1-7 to Fire Mastery (any class)
Blade of Ali Baba	Tulwar	107	+1-2 All Skills +10-30% faster cast rate
The Atlantean	Ancient Sword	105	+1 Sword Mastery
Carin Shard	Petrified Wand	107	+20-40 All Res 10% reduced damage
Blackhand key	Grave Wand	115	+7 to random necromancer skill

Andariel's Visage	Demonhead	102	+3/lvl life
Spirit Ward	Ward	105	Requirements -100% 4 sockets
Marrowwalk	Boneweave Boots	121	+1-2 necromancer skills
Cranebeak	War Spike	107	L10-20 Holy Shock Aura
Widowmaker	Ward Bow	110	+2/lvl% enhanced damage
Hellrack	Colossus Crossbow	105	-10-25% fire res -10-25% lightning res -10-25% cold res
Wizardspike	Bone Knife	118	+1-3 All Skills
Fleshripper	Fanged Knife	110	L10-20 Concentration Aura
Ghostflame	Legend Spike	107	Level 50 Venom 100 Charges Level 50 Enchant 100 Charges
Demon's Arch	Balrog Spear	107	10% L30 Burst of Speed on kill
Baranar's Star	Devil Star	118	L10-20 Conviction Aura
Schaefer's Hammer	Legendary Mallet	118	+1.5/lvl% enhanced damage
Bonehew	Ogre Axe	102	(modify)50% L30 Bone Spear on hit 50% L30 Bone Spirit on hit 50% L30 Teeth on hit
The Redeemer	Mighty Scepter	121	(modify)Damage +400-500 +7 to Smite
Ondal's Wisdom	Elder Staff	125	L20 Redemption Aura 4 sockets
Flamebellow	Balrog Blade	102	100% L40 Fire Wall on kill
Lacerator	Winged Axe	107	L15-25 Fanatism Aura
Ars Al'Diabolos	Blasphemous Grimoire	121	+3-7 to Lower Resistances
Ars Dul'Mephistos	Occult Tome	112	+1-3 to eldritch skills
Ars Tor'Baalos	Blasphemous Compendium	115	+1-5 Battle Orders +1-5 Battle Command
Gheed's Wager	Troll Belt	110	+25-50% magic find
Wraithstep	Mirrored Boots	105	+10% attack speed +10% faster cast rate
Bloodpact Shard	Mithril Point	115	+1-5 to demon skills
Dreadfang	Legend Sword	118	+300-1500 magic damage
Sling	Ring	100	10% reduced damage
Opalvein	Ring	125	+1 to random class skills

The following ones are of normally bad uniques that have some personality, widening the range of potential affixes substantially so that, in lucky cases, they present pretty unbalanced items with some very good stats and

other terrible stats. This adds a lot of variability.

Wormskull	Bone Helm	120	+1-5 Necromancer Skill Levels
Darkglow	Ring Mail	104	All Resistances +10-150
Rattlecage	Gothic Plate	110	25-100% Crushing Blow Hit Causes Monster to Flee 40-100%
Silks of the Victor	Ancient Armor	119	+1-5 to All Skills
Steelclash	Kite Shield	108	+1-4 to Paladin Skill Levels All Resistances +15-100
Goblin Toe	Light Plated Boots	106	25-100% Crushing Blow
Lenymo	Sash	110	Regenerate mana 30-150% All Resistances +5-100 +15-150 Mana
Bladebuckle	Plate Belt	102	+30-300 Defense +30-60% Faster Hit Recovery +10-50 Dexterity +5-50 Strength
The Chieftain	Battle Axe	113	+100-450% Enhanced Damage Adds 1-(40-500 lightning Damage)
Humongous	Giant Axe	125	+80-350% Enhanced Damage Adds 8-(15-150) Damage 33-99% Crushing Blow Requirements -20-100% +20-100 Strength
Hellclap	Short War Bow	112	Adds 15-(30-500) Fire Damage +1-7 Fire Skills
The General's Tan Do Li Ga	Flail	116	+50-450% Enhanced Damage Adds 1-(20-200) Damage 20-60% Increased Attack Speed
The Grim Reaper	War Scythe	109	+20-450% Enhanced Damage +15-150 Minimum Damage
Spire of Lazarus	Gnarled Staff	121	+1-5 Sorceress Skill Levels +2-7 Lightning +1-7 Chain Lightning
The Salamander	Battle Staff	121	+2-5 Fire Skills +2-7 Fire Ball +1-7 Fire Wall

Blood Crescent	Scimitar	125	+60-450% Enhanced Damage 15-60% Increased Attack Speed 15-50% Life Steal All Resistances +15-75 +15-150 Life
Crow Caw	Tigulated Mail	108	15-60% Increased Attack Speed 15-60% Faster Hit Recovery +15-50 Dexterity
Tiamat's Rebuke	Dragon Shield	104	27-(53-350) Cold Damage 35-(95-350) Fire Damage 1-(120-450) Lightning Damage All Resistances 25-75
Witchwild String	Short Siege Bow	122	+150-450% Enhanced Damage 2-5 Sockets
Langer Briser	Arbalest	114	+170-350% Enhanced Damage +10-150 Maximum Damage 30-300% Better Chance of Getting Magic Items
Bloodtree Stump	War Club	117	+180-450% Enhanced Damage +25-125 Strength +2-7 to Masteries +3-7 to Mace Mastery
The Meat Scraper	Lochaber Axe	111	+150-450% Enchanced Damage 30-60% Increased Attack Speed 10-30% Life Steal 25-250% Magic Find
Hand of Blessed Light	Divine Scepter	109	+2-5 Paladin +2-7 Fist of the Heavens
Kelpie Snare	Fuscina	117	+140-350% Enhanced Damage Adds 30-(50-200) Damage +10-30 Strength
Razorswitch	Jo Staff	123	+1-7 All Skills +30-60% Faster Cast Rate All Resistances +50-100 +80-175 Life
Bloodletter	Gladius	118	+140-450% Enhanced Damage Adds 12-(45-200 Damage) +90-300 Attack Rating 20-60% Increased Attack Speed 8-25% Life Steal +1-7 to Whirlwind +2-7 to Blade Mastery

Swordguard	Executioner Sword	120	+170-350% Enhanced Damage Requirements -50-100% All Resistances +10-100 20-60% Faster Hit Recovery 20-60% Increased Chance of Blocking
Suicide Branch	Burnt Wand	113	+1-5 to All Skills Increase Maximum Mana 10-100% All Resistances +10-50 +40-100 Life
Veil of Steel	Spired Helm	124	+60-400% Enhanced Defense +140-1000 Defense All Resistances +50-100 +15-75 Strength +15-75 Vitality
Giant Skull	Bone Visage	101	10-100% Crushing Blow +25-100 Strength 1-3 Sockets
Templar's Might	Sacred Armor	125	+10-100 Strength +10-100 Vitality +20-60% Faster Hit Recovery +1-7 Offensive Auras
Nosferatu's Coil	Vampirefang Belt	111	Slows Target by 10-100% 5-50% Life Steal +15-50 Strength 10-60% Increased Attack Speed
Arioc's Needle	Hyperion Spear	113	+180-450% Enhanced Damage +30-60% Increased Attack Speed +2-7 All Skills
Lightsabre	Phase Blade	117	+150-450% Enhanced Damage Adds 10-(30-200) Damage Adds 60-(120-300) Magic Damage Adds 1-(200-400) Lightning Damage 5-50% Mana Steal 20-60% Increased Attack Speed

16.2 Super high tier uniques

These can only be dropped by heralds of certain tiers. These are the modifications we make to heralds, apart from increasing their chance of uniques:

Tier	Damage boost	Health boost
1-2	25	500
3-4	50	1000
5-6	75	1500
7-8	100	2000
9-10	125	2500
11-12	150	3000
13-14	175	3500
15-16	200	4000
17-18	250	5000
19-20	300	6000
21-22	350	7000
23-24	400	8000
25-26	450	9000
27	500	10000

And these are the very high level uniques, the herald tier required to drop them, and the chance (per item dropped) that they drop them.

Name	Base item	Tier	Chance (1 in..)	Properties
Obsidian Ring of the Zodiac	Ring	6	500	+1 All Skills +10% Faster Cast Rate +20% Maximum Mana +5-15 All Resistances +5-10% Life Leech +5-10% Mana Leech Cannot be Frozen L10-15 Meditation Aura
Damned Halo	Tiara	13	750	+2 All Skills 3 Sockets +40-75 All Resistances +5 All Max Resistances 25-50% Reduced Damage +10-25% Life Leech +10-25% Mana Leech Cannot be Frozen 25% L10 Mind Blast on hit
The Cloak of Abbadon	Dusk Shroud	20	1000	+4 All Skills 4 Sockets +100-150 All Resistances +10 All Max Resistances 25-50% Reduced Damage +50% Faster Cast Rate +500% Enhanced Damage 50% Crushing Blow Cannot be Frozen +5-10 Teleport
Soul of Acheron	Jewel	27	1500	+2 All Skills +10% Fire Skill Damage +10% Cold Skill Damage +10% Lightning Skill Damage +10% Poison Skill Damage +50% Increased Attack Speed +15% Faster Cast Rate +75% Enhanced Damage +15% Maximum Life +30% Faster Hit Recovery +150-300% Gold Find

17 Unique mutations

In order to make new unique drops more interesting when we already have a supply of them, we add a possibility to mutate uniques. This gives them a certain significant negative attribute, and a possibility of a large positive attribute. This means that most of the time the unique becomes worse and therefore is likely to be discarded, but occasionally it becomes better. There is no limit on applying these, but trying to get more than 1 is definitely a risky move.

Item types	Negative mutation	Positive mutation	Chance	Recipe
Armor	-5-15 vitality	20% L15 Cyclone Armor when hit	4%	3 UE + Perfect Amethyst
		20% L15 Bone Armor when hit	4%	
		L15-30 Cleansing Aura	4%	
		+1-20 Natural Resistance	1%	
Armor	-5-15 energy	+75-250% magic find	4%	3 UE + Perfect Diamond
		+200-500% gold find	4%	
		+5-20 Find Item	4%	
		+5-10 Teleport	1%	
Armor	-20-50 fire res	+500-2500 defense	4%	3 UE + Perfect Ruby
		Damage reduced 10-30%	4%	
		+2-15 all max res	4%	
		2-10% all absorbs	1%	
Armor	-20-50 lightning res	+500-2500 defense	4%	3 UE + Perfect Topaz
		Damage reduced 10-30%	4%	
		+2-15 all max res	4%	
		2-10% all absorbs	1%	
Armor	-20-50 cold res	+500-2500 defense	4%	3 UE + Perfect Topaz
		Damage reduced 10-30%	4%	
		+2-15 all max res	4%	
		2-10% all absorbs	1%	
Armor	-20-50 poison res	+500-2500 defense	4%	3 UE + Perfect Topaz
		Damage reduced 10-30%	4%	
		+2-15 all max res	4%	
		2-10% all absorbs	1%	

Weapon	Physical damage increased by 5-10%	+1-2 all skills	4%	3 UE + El
		20-60% faster cast rate	4%	
		L10-30 Meditation Aura	4%	
		L15-30 Conviction Aura	1%	
Weapon	-5-15 strength	+15-50% fire damage	4%	3 UE + Ral
		-15-50% enemy fire resistance	4%	
		+1-20 Lower Resist	4%	
		Sunder Fire Immunity	1%	
Weapon	-5-15 strength	+15-50% lightning damage	4%	3 UE + Ort
		-15-50% enemy lightning resistance	4%	
		+1-20 Lower Resist	4%	
		Sunder Lightning Immunity	1%	
Weapon	-5-15 strength	+15-50% cold damage	4%	3 UE + Thul
		-15-50% enemy cold resistance	4%	
		+1-20 Lower Resist	4%	
		Sunder Cold Immunity	1%	
Weapon	-5-15 strength	+15-50% poison damage	4%	3 UE + Tal
		-15-50% enemy poison resistance	4%	
		+1-20 Lower Resist	4%	
		Sunder Poison Immunity	1%	
Weapon	-5-15 dexterity	Sunder Physical Immunity	4%	3 UE + Eld
		75-300% bonus to attack rating	4%	
		15-75% crushing blow	4%	
		L10-30 Fanaticism Aura	1%	

18 Per Character special items

One issue with this game sometimes is that a character easily becomes irrelevant when another character does everything better than it. Building them for the sake of building them is fun, but it feels like it's hard to tell when you feel you've done enough with that character. To deal with this in a gamified way, we add a new unique quest item, Pure Hellfire, that drops only at level 120 or higher, that cannot be transfered between characters and that you can only have one of. A character is not considered *complete* until it has one. This item is designed to replace the tome of town portal, and as such gives the following attributes:

- +1 to Town Portal
- +10% experience gained
- +100% gold dropped from monsters

- 5% reduced vendor prices

19 Not added to this document yet

20 Ideas not explored yet

- More ethereal uniques / unique versions - Every single one should be special.
  - Charms - Leave for now.
  - Rings and amulets - Leave for now.
  - Powerful ethereal gloves, belts, boots for unused base types.
  - Ethereal weapons and shields with charges of powerful one time skills (enchant, thunder storm, blade shield, energy shield, cold armours, holy shield, battle orders, burst of speed, fade, venom, bear, shadow master, hurricane, armageddon, hex purge, goat-man, tainted, defiler, consume).
  - Ethereal powerful items on bases with only 1 socket max (daggers, buckler).
  - Cast summons on/when hit - persistence (valkyrie, golems, spirits, wolves, bear, vines, shadow master, defiler).
  - Huge oskills on otherwise mediocre items.
  - Specific weapon masteries, especially 2-hand.
  - Fire/cold/lightning mastery on big 2-handed weapons with hybrid gameplay of some kind.
  - Oskill useful skills (telekinesis, valkyrie, golems, wolves, bear, shadow master, defiler, bone armor, cyclone armor, blade shield, cleansing).
  - Cast teleport, taunt, psychic hammer when hit.
  - Huge absorption.
  - Big negative properties (see above).
  - Pierce immunities.
  - 100+ versions of bad uniques that just have wider ranges on attributes (big potential, requires luck).
- More and more interesting mutations that use organs.

- Temporary charges on charms from combination recipes.
- Iron Golem item by above 100 level requirement
- Casting monster abilities
- All class-specific runewords
- Chaos Tristram Terror Zone.
- Vendor that sells all types of jewelry for very expensive prices.
- Items that drop from specific monsters and can only be used within the same game, to push the idea of doing multiple areas in the same game.
- Play around with ItemTgtDo on skills (when item casts it, the monster does it rather than the player). Think about funny skills with this. Teleport was like this, what about blaze/thunderstorm, what about others?
- Consider very long term character-instransferable drops from high heralds that give permanent bonuses. Very slow but indefinite progression. Either on the character or on the Pure Hellfire (e.g. +skills, +stats, +life, +max res...)